

Results: Heat Exposure and Cooling Efficiency at Oregon Data Centers

The 109 Oregon data centers identified in the facility inventory are concentrated in five of Oregon's nine Level III ecoregions (Omernik 1987): Columbia Plateau (61 facilities), Willamette Valley (33), Blue Mountains (13), Eastern Cascades Slopes and Foothills (1), and Klamath Mountains/California High North Coast Range (1). The inventory records current facility locations only and does not identify planned or future sites. The emphasis placed on Willamette Valley in the sections below reflects its standing as the second-largest current cluster and its projected trajectory, not an assumption about where future facilities will be built.

Current heat exposure and where change concentrates

Columbia Plateau is already the most heat-exposed region in the state (Figure 1, top). Its historical average of threshold-exceedance days, days on which conditions exceed the operating range for which data center cooling equipment is designed, is roughly four to five times that of Willamette Valley. Willamette Valley is, at present, among the mildest of the five regions by this measure.

Facilities in Willamette Valley also show a wider ensemble spread in projected exceedance days than facilities in Columbia Plateau or Blue Mountains: a relative spread, the ensemble's 5th–95th percentile range divided by its mean, of 0.40, compared with 0.24 and 0.28 in those two regions. This pattern is consistent with Willamette Valley's low historical exceedance-day count sitting close to the threshold itself, where small differences among global climate models can shift the projected day count by a large fraction of a small baseline.

This current-exposure ranking is inverted by end-of-century, in all three metrics evaluated (Figure 1, bottom). Willamette Valley's representative facility shows an end-of-century change factor of 6.3 for threshold-exceedance days, 1.033 for PUE, and 3.4 for WUE; Blue Mountains shows a similar pattern, at 6.9, 1.016, and 2.9 respectively. Columbia Plateau's representative facility, by contrast, shows the smallest change factor of the three multi-facility regions on every metric: 3.3, 1.023, and 2.0. The two regions with the mildest current climate are projected to see the largest proportional change in every metric evaluated, while the region with the most facilities today and the highest current exposure is projected to see the smallest proportional change. The mechanism behind this pattern, and its implications for facility-level siting decisions, is developed further below.

Figure 1. Historical heat exposure (top) and end-of-century change factor by metric (bottom)
Ecoregions shaded by their representative facility's value; points show every facility

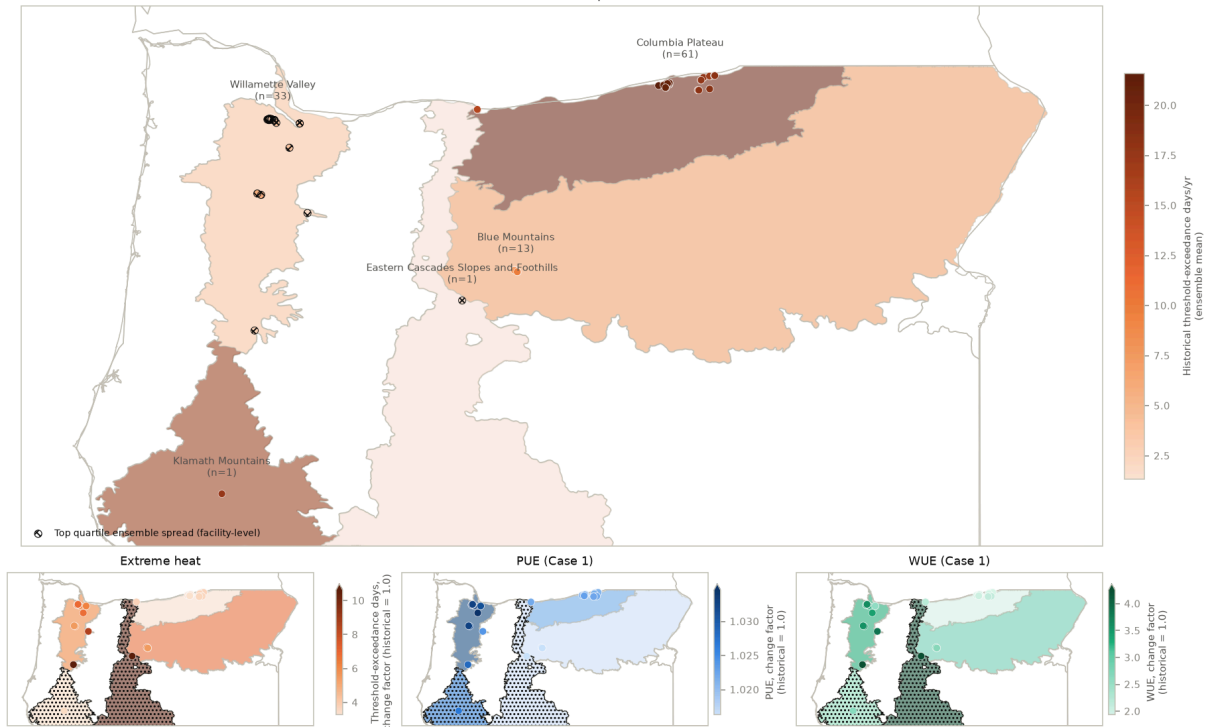


Figure 1. Top: Oregon data center locations and historical heat exposure, in threshold-exceedance days per year (ensemble mean). Bottom: end-of-century change factor for threshold-exceedance days, PUE, and WUE, relative to each facility's own historical baseline (historical = 1.0). Ecoregions are shaded by their representative facility's value; all facilities are plotted as points. Cross-hatching in the top panel flags facilities in the top quartile of relative ensemble spread; cross-hatching in the bottom panels flags Eastern Cascades Slopes and Foothills and Klamath Mountains/California High North Coast Range, which are excluded from each panel's color scale because their near-zero historical baselines produce disproportionately large change factors (see Limitations).

Divergence between heat exposure and cooling response

Power Usage Effectiveness (PUE), the ratio of a data center's total electricity use to the electricity used by its computing equipment alone, and Water Usage Effectiveness (WUE), the equivalent measure for water, are both projected to rise as cooling becomes less efficient under warmer, more humid conditions. Statewide fleet-mean heat exposure, PUE, and WUE respond very differently to the same projected warming, expressed here as each metric's own change factor relative to its historical baseline (Figure 2). Threshold-exceedance days are projected to increase from a historical average of 13 days per year (5th–95th percentile: 3–22) to 53 days per year (20–86) by end-of-century, a change factor of roughly 4.0. WUE is projected to increase from 0.25 liters per kilowatt-hour (0.12–0.34) to 0.60 (0.32–0.80), a change factor of roughly 2.4. PUE is projected to increase from 1.134 (1.127–1.140) to 1.162 (1.144–1.177), a change factor of roughly 1.03.

Ensemble uncertainty differs accordingly. The projected range for exceedance days is wide relative to its own mean; the range for PUE is narrow; the range for WUE falls between the two. Data center cooling systems are likely to continue operating through projected warming at a modest efficiency cost, while the water required to operate them is likely to increase in closer proportion to the increase in heat exposure itself. Efficiency and water demand should therefore be treated as distinct risks rather than a single undifferentiated measure of cooling-system stress.

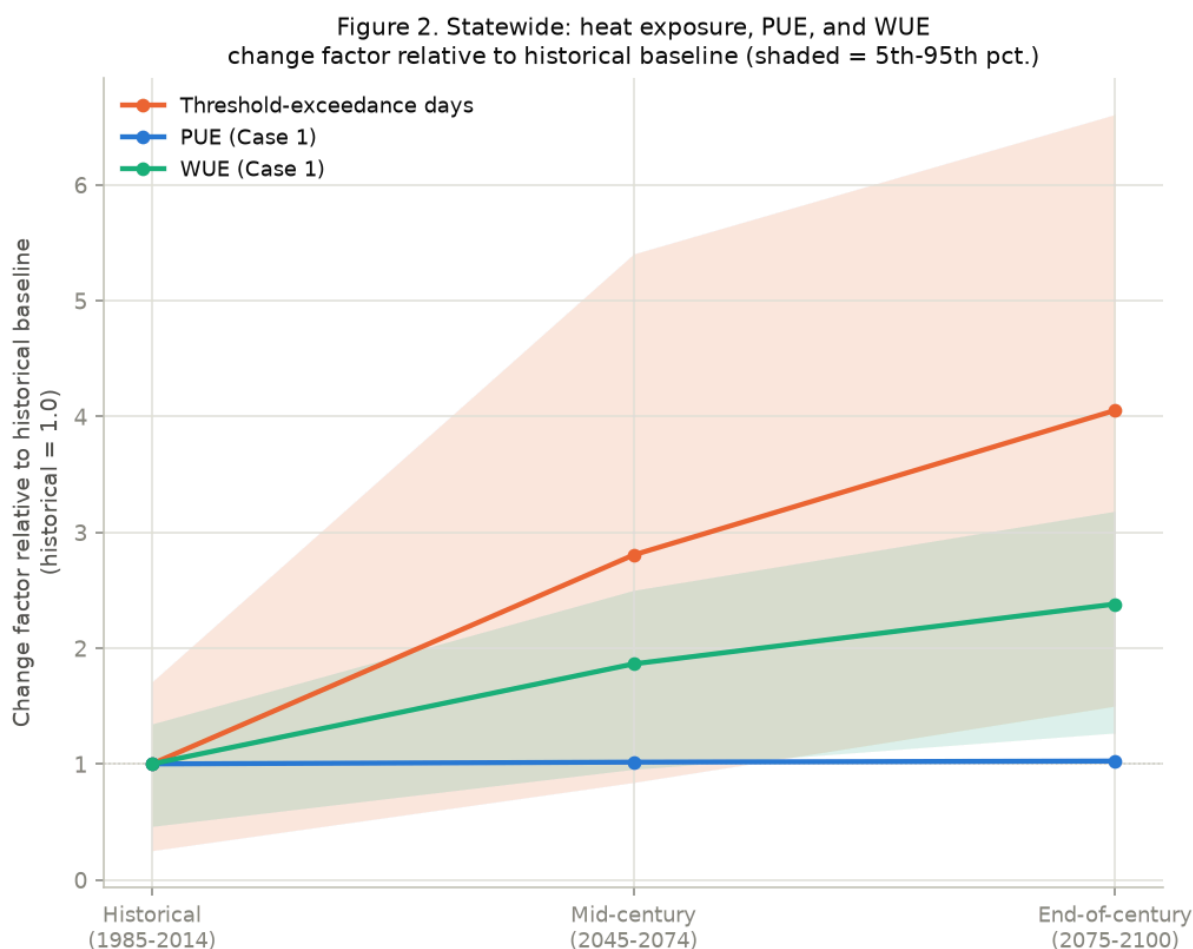


Figure 2. Statewide fleet-mean heat exposure, PUE, and WUE, each expressed as a change factor relative to its own historical baseline (historical = 1.0). Shaded bands show the 5th–95th percentile range across the climate model ensemble.

Seasonal timing and cooling-system design

Nearly all of the projected change in PUE and WUE is concentrated in a five-month period from May through September, with little change projected in the remaining months (Figure 3). Cooling-system design affects both the size and the shape of this seasonal response. Large-scale systems combining an airside economizer with adiabatic cooling show the largest seasonal increase of the three archetypes

evaluated, reaching a projected September PUE increase of 0.07 and a WUE increase of approximately 0.9 liters per kilowatt-hour. Midsize systems without an economizer show a smaller peak in August in both metrics. Large-scale systems using a waterside economizer show the smallest fleet-wide seasonal increase of the three archetypes.

Figure 3. Seasonal shape of the response, by archetype

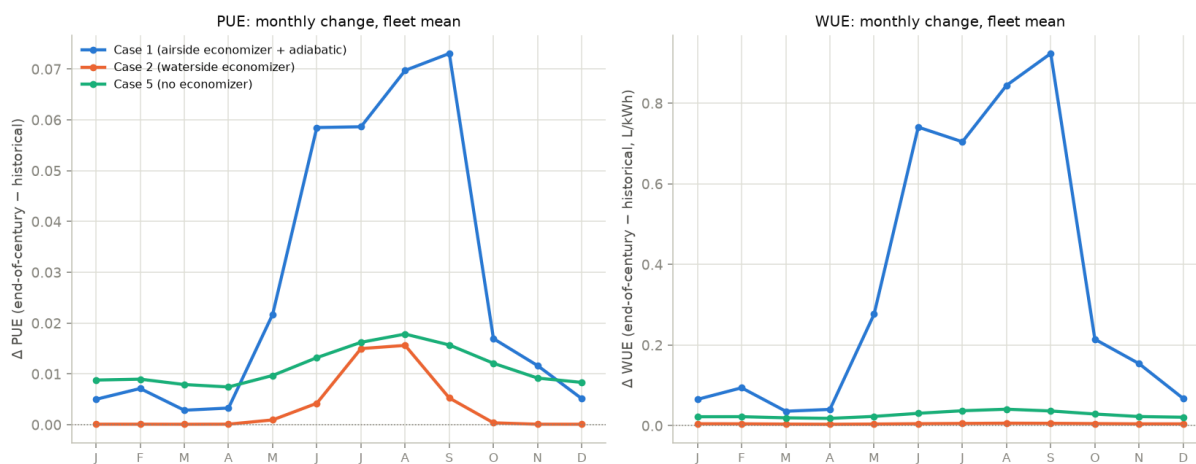


Figure 3. Fleet-mean change in monthly PUE and WUE, end-of-century minus historical, by cooling-system archetype.

Facility-level variation and siting implications

Across the 109 facilities, historical cooling degree days are negatively correlated with the projected absolute increase in both PUE (Pearson $r = -0.33$) and WUE ($r = -0.36$) (Figure 4). Facilities in currently milder locations are projected to see a larger absolute increase in both metrics than facilities that are already hot. The similarity in strength and direction between the two correlations indicates a general property of the cooling response rather than an effect specific to either metric.

A plausible mechanism follows from how economizer-based cooling operates. These systems draw on outside air for cooling whenever ambient conditions allow it, switching to more energy- and water-intensive mechanical or adiabatic cooling once conditions exceed a threshold. A facility in a currently mild climate spends relatively few days per year above that threshold today, leaving more days available to newly exceed it as the climate warms. A facility already in a hot climate spends much of the year in mechanical-cooling mode already, leaving less room to change, even as its total heat exposure remains higher and continues to grow fastest in absolute terms.

Willamette Valley is already the second-largest data center cluster in Oregon and, as shown above, is one of the two multi-facility regions where this pattern is most pronounced. Sites in currently cooler parts of Oregon are exposed to less absolute heat, but current climate alone does not proportionally reduce future cooling-efficiency or water-use risk. Adaptation planning that treats current climate as the

primary siting criterion may understate future operating risk for a region that already hosts a substantial share of the state's data center fleet.

Figure 4. Facility-level: hotter-today sites see smaller absolute degradation
(n=109 facilities; hollow markers = single-facility ecoregions)



Figure 4. Facility-level historical cooling degree days versus facility-level absolute change in PUE (left) and WUE (right), historical to end-of-century, colored by ecoregion. Hollow markers denote the two single-facility ecoregions.

Limitations

Facility-level correlations in this analysis use absolute change rather than percent change. Historical WUE is strongly correlated with current heat exposure, so normalizing by each facility's own historical value would substantially inflate the apparent strength of its correlation with heat exposure, independent of the underlying relationship being measured. Absolute-value comparisons avoid this effect and are used throughout Figure 4.

Eastern Cascades Slopes and Foothills and Klamath Mountains/California High North Coast Range are each represented by a single facility in the inventory. Results for these two ecoregions are shown throughout this analysis but should be interpreted with caution given the limited sample, and historical values near zero in these regions can produce large proportional changes that are not necessarily representative of a broader regional pattern. In Figure 1's change-factor panels, these two ecoregions are excluded from the color scale for this reason and are marked with cross-hatching rather than a fill color.